HappyClassify & HappyPredict

Machine Learning-Based Classification of World Happiness Levels and Comparative Analysis of Machine Learning Regression Models for Predicting World Happiness Scores

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Overview

- Project overview & app architecture — one unified Streamlit app, two dashboards
- The dataset — 165 countries, 2005–2023, 8 socio-economic features
- **HappyClassify** — EDA, country explorer, model comparison (Random Forest ∼83% accuracy), live demo
- **HappyPredict** — EDA, model comparison (MAE / RMSE / R^2), live demo
- Conclusion & future scope

World Happiness ML Suite — One Unified Web App

app.py routes between two purpose-built dashboards using Streamlit session state.

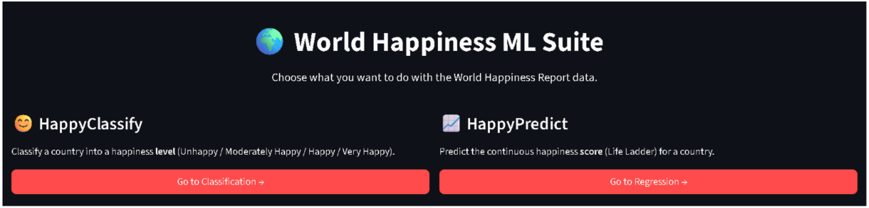
HappyClassify

- Classifies a country into a happiness tier: Unhappy, Moderately Happy, Happy, or Very Happy

HappyPredict

- Predicts the exact continuous happiness score (Life Ladder) for a country

Built with a shared data pipeline, a shared visual theme, and deployed live on Streamlit Community Cloud.

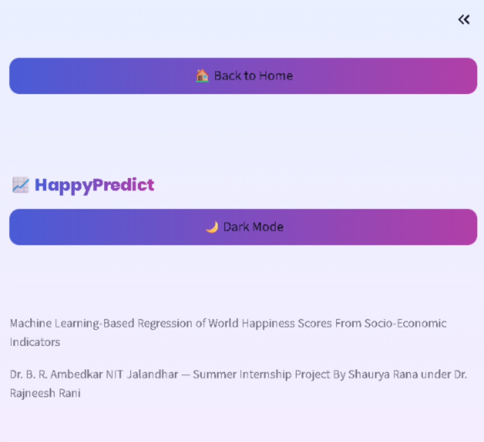


App Architecture

How It's Wired Together

- `app.py` — router, using `st.session_state`
- `classify_page.py` — `render_classify()`
- `regression_page.py` — `render_regression()`

- Home page → two clickable cards call `go("classify")` / `go("regress")`
- Sidebar → "Back to Home" button appears once inside either dashboard
- Rerun model → Streamlit reruns the whole script on every click; `session_state` persists the page choice



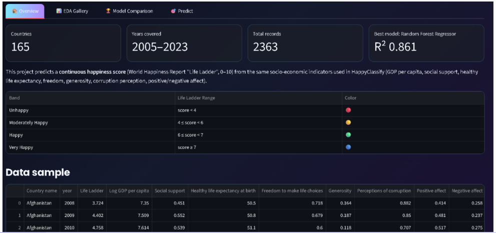
The Dataset

Foundation

Socio-economic input features (8):

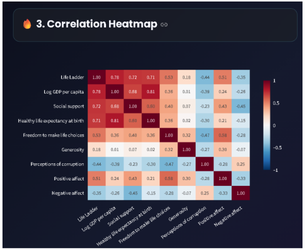
- Log GDP per capita
- Social support
- Healthy life expectancy
- Freedom to make life choices
- Generosity
- Perceptions of corruption
- Positive affect
- Negative affect

- **165** countries covered
- **2005–2023** years of WHR panel data
- **8** socio-economic input features



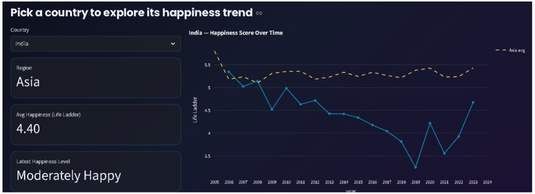
HappyClassify — Overview & Exploratory Data Analysis

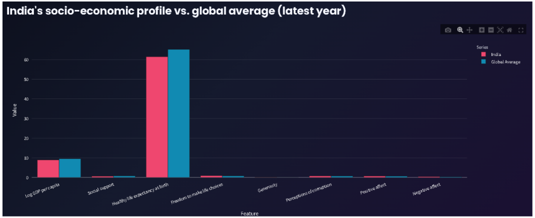
- 4 happiness tiers engineered from the continuous Life Ladder score
- 9-chart EDA Gallery — histograms, correlation heatmap, world map, radar chart, and more
- Cached with `@st.cache_data` so charts don't recompute on every click



HappyClassify — Country Explorer

- Pick any country and see its happiness trend over time
- Compared against its regional average
- Feature comparison chart showing how the country stacks up across all 8 indicators





HappyClassify — Model Comparison

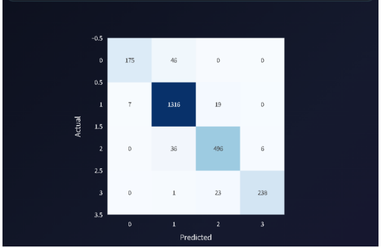
Models trained:

- Logistic Regression
- KNN
- Decision Tree
- **Random Forest** ★
- SVM

- Random Forest was the best performer — ∼**83% test accuracy**
- Per-model diagnostics: confusion matrix, ROC curve, feature importance, class distribution

Model Comparison Table

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Training Time (s)
Random Forest	0.837	0.835	0.724	0.763	0.954	1.500
KNN	0.800	0.741	0.697	0.713	0.936	0.007
SVM	0.766	0.749	0.632	0.664	0.924	0.550
Logistic Regression	0.738	0.681	0.610	0.634	0.920	0.033
Decision Tree	0.758	0.709	0.670	0.684	0.864	0.019

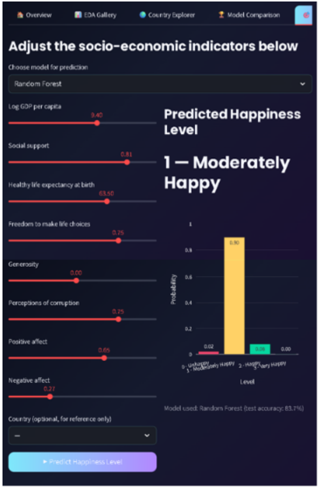


A. Confusion Matrix (Random Forest)

HappyClassify — Demo

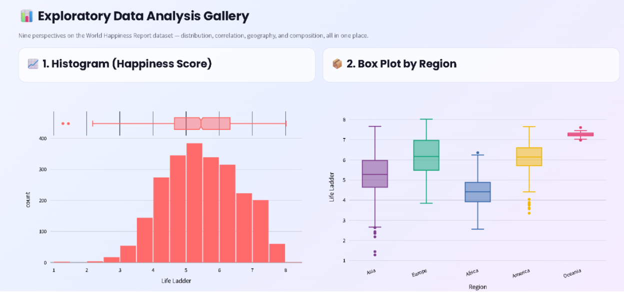
Live Prediction

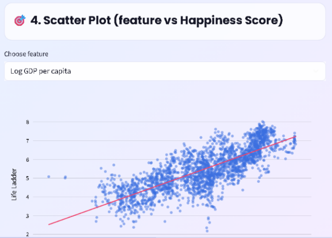
- Sliders for all 8 socio-economic features
- Model selector to pick which of the 6 trained classifiers runs the prediction
- Instant output: predicted happiness tier + a probability breakdown bar chart



HappyPredict — Overview & Exploratory Data Analysis

- Same structure as HappyClassify, focused on regression instead of classification
- 4 models trained: Linear Regression, Decision Tree, Random Forest, SVR
- Target: the continuous Life Ladder score (0–10), not a discrete tier





HappyPredict — Model Comparison & Diagnostics

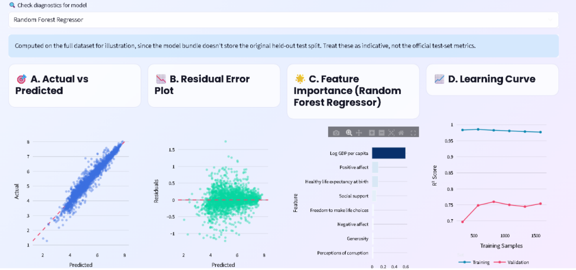
Metrics used:

- MAE — Mean Absolute Error
- RMSE — Root Mean Squared Error
- R^2 — Coefficient of Determination

- Learning curves per model to assess bias–variance trade-offs
- Ensemble methods (Random Forest) generally outperformed linear/SVR on this data

Model Comparison Table

Model	MAE	MSE	RMSE	R2	Training Time (s)
Random Forest Regressor	0.305	0.166	0.407	0.861	3.039
Decision Tree Regressor	0.390	0.270	0.519	0.774	0.015
SVR	0.339	0.193	0.440	0.838	0.157
Linear Regression	0.415	0.283	0.532	0.763	0.059



- Adjust sliders for GDP, social support, life expectancy, and more
- Choose a model — here, Random Forest Regressor
- Instant output: predicted Life Ladder score shown as a number and a gauge chart, with its happiness label

Conclusion & Future Scope

- ML classification effectively distinguishes world happiness levels
- Regression models compared for accuracy in predicting scores
- Key insight: best model is Random Forest in HappyClassify, and Random Forest Regressor in HappyPredict
- Future scope: integrate larger datasets, advanced deep learning, and real-time happiness monitoring

Thank You